Generating Large Primes Efficiently

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Part I

The Details

1 Introduction

1.1 RSA Cryptography

Definition RSA cryptography is a public-key encryption system widely used to protect digital communication. It relies on a pair of mathematically linked keys: a public key, which can be shared, and a private key, which must be kept secret. RSA key generation begins by selecting two distinct, large random prime numbers, conventionally called p and q. Their product, $n = pq$, forms part of the public key. The algorithm also uses an exponent chosen to work with the primes; the corresponding private exponent is calculated using number-theoretic relationships. A message can then be encrypted with the public key and recovered with the private key. RSA can also be used for digital signatures.

Its security depends on the practical difficulty of factoring the product of the two primes. Therefore, the primes must be generated using a cryptographically secure random source and tested

for primality. In real systems, RSA is generally used with secure padding and often protects a key that is then used by a faster symmetric cipher.

For a 2048-bit RSA modulus, the two primes are typically about 1024 bits each, or roughly 300 decimal digits per prime. A 3072-bit modulus uses primes of about 1536 bits each. The modulus size refers to the combined bit length of the product, not the size of either prime alone. Larger keys generally provide a greater security margin but require more computation. Prime generation must therefore balance randomness, reliable primality testing, and the intended security level.

2 Our approach

How we want to generate primes

The is approach exploits the fact that all primes end in 1,3,7,9. Taking a number R as Root, we add the digits 1 or 3 or 7 or 9 at the end. These new generated numbers are called branches. Then check each approach for it's primality.

For E.g Let us take R = 1, the branches of the number 1 are 11,13,17,19. After checking each number for it's primality we find that all branches are prime. Then we each number as a new root, thus we get the following numbers

For R = 11 we get 111, 113, 117, 119

For R = 13 we get 131 ,133, 137, 139

For R = 17 we get 171, 173, 177, 179

For R = 19 we get 191, 193, 197, 199

From this the non primes are 111,117,133,171,177. Now as we can see , only a few branches are prime. However this is dependent on the root number, we need to get a sufficiently large enough root number that cannot be deciphered. For e.g using 201 we get results such as 2011,2013,2017,2019 out of which only 2011 and 2017 are prime. Thus a good root will have

many branches that are prime.